

AIDA TARIGHAT

Computational Psycholinguistics and Human–AI Interaction

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I go by and publish under Aida Tarighat; official records list my full name as Fatemeh Samadzadeh Tarighat.

RESEARCH PROFILE

Interdisciplinary researcher working across applied linguistics, psychology, and human–computer interaction, with a sustained focus on human–AI communication: how people interpret, trust, and respond to language and social cues produced by artificial agents. Background combines behavioral experimentation, corpus analysis, computational modeling, and participatory/co-design methods, built on a trajectory that moved deliberately from literature and language, to psychology, to computational methods in order to keep pace with the same underlying questions about language, meaning, and communication technology.

Research Focus

- Predictive processing and human–AI behavioral divergence: how humans and artificial agents handle disfluency and nonliteral meaning in multimodal settings
- Human-centered, context-aware intelligent systems: improving interpretability and human–AI interaction quality
- Cross-linguistic corpus analysis and computational modeling of language structure and category learning
- Trust, deception, and manipulation via psycholinguistic cues (e.g., hesitation markers, hedging language) in human evaluation of AI-generated content
- Emotion regulation and social-psychological responses to adversity

Methods and Tools

- Behavioral experimentation: eye-tracking, self-paced reading, cloze completion, survey design, interviews
- Statistical modeling in R: regression, Bayesian inference, multivariate methods, permutation-based analyses
- Language model evaluation in Python (PyTorch, Hugging Face, Pandas): MLM/CLM paradigms, LSA cosine similarity, Shannon entropy, cross-tokenization alignment, Jensen-Shannon divergence
- Parameter-efficient fine-tuning: LoRA, prompt tuning

Applications

- Human-AI dialogue and conversational AI design
- Human-computer interaction, co-design, and human-centered system design
- Computational modeling of language and cross-linguistic corpus analysis
- AI safety, societal risk evaluation, and accessible, assistive, educational, and e-mental health technologies.

EDUCATION

Ph.D. Natural Language Processing — *University of Edinburgh, UK* — 09/2021–06/2026

- Thesis: Written Disfluencies and Nonliteral Meaning in Natural Language Processing: Evidence from Behavioral and Computational Studies
- Supervisors: Martin Corley, Patrick Sturt

M.Sc. General Psychology — *University of Tabriz, Iran* — 09/2018–04/2021

- Thesis: Comparing the Aspects of Emotional Regulation in Individuals With and Without Sexual Traumatic Experiences
- Supervisors: Mostafa Zarean, Darejan Jana Javakhishvili

M.A. English Language and Literature — *Shahid Chamran University, Iran* — 09/2010–09/2013

- Thesis: A Social Psychological Analysis of Power Distribution in José Saramago’s Blindness (1997 Pontiero translation)
- Supervisors: Mahmoud Daram, Saleh Hosseini

B.A. English Language and Literature — *Shahid Chamran University, Iran* — 09/2006–07/2010

Career break, 2013–2018, for full-time caregiving.

PUBLICATIONS

Peer-Reviewed Conference Papers

- **Tarighat, A.**, Sturt, P., Corley, M. (2024). Perspectives on Language Model and Human Handling of Written Disfluency and Nonliteral Meaning. Proc. 28th Workshop on the Semantics and Pragmatics of Dialogue (Trentologue).
- Hew, J., **Tarighat, A.**, Corley, M., Horne, Z. (2024). Examining structural and semantic predictors of announced sarcasm on r/AskReddit. Proc. Annual Meeting of the Cognitive Science Society, pp. 3364–3370.
- **Tarighat, F. S.**, Magdy, W., Corley, M. (2022). Understanding Fillers May Facilitate Automatic Sarcasm Comprehension. Proc. 26th Workshop on the Semantics and Pragmatics of Dialogue (DubDial), pp. 215–217. (*published as Fatemeh S. Tarighat*)
- **Tarighat, F. S.**, Zarean, M. (2021). Cognitive Emotion Regulation in Individuals With and Without Sexual Traumatic Experiences. National Congress of Family Psychology Association of Iran. (peer-reviewed; Persian; no paginated proceedings) (*published as Fatemeh S. Tarighat*)
- Zarean, M., **Tarighat, F. S.**, Clough, B. (2019). Preferences and attitudes toward e-mental health services among Iranians. 1st Biennial Congress on Psychology and Behavioral Sciences. (peer-reviewed; Persian; no paginated proceedings) (*published as Fatemeh S. Tarighat*)

Peer-Reviewed Journal Articles in Preparation

- **Tarighat, A.**, Sturt, P., Corley, M. Umm, What Comes Next? Human and LM Predictions Under Written Filler.
- **Tarighat, A.**, Sturt, P., Corley, M. It’s Um Hard to Process but Not Hard to Evaluate: Written Fillers and Sarcasm in Reading and Evaluation Experiments. Target: Discourse Processes.

- **Tarighat, A.**, Zarean, M., Javakhishvili, J. Childhood Sexual Trauma Selectively Disrupts Cognitive and Interpersonal Emotion Regulation in Adults. Target: European Journal of Psychotraumatology.
- **Tarighat, A.**, Zarean, M. Contemporary Directions in Emotion Management: A Conceptual Review. Target: Motivation and Emotion.

Book Chapter

- Zarean, M., **Tarighat, F. S.** (2022). Hearing the Untold: A Review of Central Asia’s Contribution to the Expansion of Psychological Assessment. Comparative Histories of Psychological Assessment. Cambridge University Press. *(published as Fatemeh S. Tarighat)*

Abstracts

- **Tarighat, A.**, Sturt, P., Corley, M. (2025). It’s Umm Hard to Process but Not Hard to Evaluate. British Psychological Society – Scottish Branch.
- Kapron-King, A., Fletcher, L., **Tarighat, A.**, Droop, S., Dobрева, R., Cummins, C., Rohde, H. (2024). Tell It Like It (Usually) Is(n’t): Speakers’ Mention of Instruments in Reddit Corpus Reflects Instrument (A)typicality. RAILS.
- **Tarighat, A.**, Corley, M. (2023). When Speech Becomes Writing: The Case of Disfluencies. CLASP Conference on Learning with Small Data.
- **Tarighat, F. S.** (2019). A Social Psychological Study of Power Relations Considering Neutral Agents and Gendered Bodies in José Saramago’s Blindness. 1st Biennial Congress on Psychology and Behavioral Sciences. *(published as Fatemeh S. Tarighat)*

GRANT PROPOSAL WRITING

Wellcome Early-Career Awards Proposal — *Sole applicant/PI — 2026*

- Authored a 5-year funding proposal on Incremental Disfluency Detection in Trauma-Informed Conversational AI (DisTAI), including a participatory co-design plan for developing and evaluating conversational-AI tools with trauma-affected participants *(application withdrawn by Wellcome)*

PROFESSIONAL EXPERIENCE

Ph.D. Researcher — *University of Edinburgh, UK — 2021–2026*

- Independently designed and led a multi-year program of mixed-methods research into an area with no established methodology, combining behavioral experiments and computational modeling to build new frameworks for understanding where human and AI language processing diverge.
- Synthesized evidence across psychology, linguistics and computer science into coherent, actionable findings, publishing and presenting across NLP, psycholinguistics and cognitive science venues.
- Communicated complex, technical findings to varied academic and interdisciplinary audiences through invited talks, conference presentations and dissemination outputs (selected list below).

Graduate Research Assistant, Department of Psychology — *University of Edinburgh, UK — 2023–2026*

- Supervised and mentored undergraduate students on psycholinguistics research projects, providing guidance on research design, data collection, and analysis.

Research Collaborator — *Swansea University, UK — 2025–Present*

- Partnering with an external academic collaborator to examine the universality of a linguistic phenomenon using a multilingual corpus; jointly devising shared labeling and evaluation protocols across languages and institutions — work requiring alignment of methods and priorities across teams.

Facilitation and Learning Design

- Designed and co-led a 4-day residential writing retreat for doctoral researchers (2024), structuring the program around peer dialogue, reflection, and collaborative problem-solving.
- Co-organized two multi-day interdisciplinary retreats (2022–2023) integrating structured reflection and shared learning for a postgraduate research cohort.
- Organized and facilitated a community wellbeing event for early-career researchers (2023).

Graduate Assistant, Department of Psychology — *University of Tabriz, Iran — 2018–2021*

- Translated and adapted standardized psychological questionnaires for use in graduate research projects.

Freelance English Language Instructor, In-Person/Online Courses — *2006–2021 (Senior Instructor, 2008–2021)*

Freelance Scientific and Literary Translator and Editor, English, Persian, Turkish, French — *2008–Present*

International Affairs Manager — *Mashal Tarigh Tejarat Aria Trading Company, Urmia, Iran — 2009–2010*

- Managed international partner communications, contractual records, and deadline tracking across multiple concurrent trade agreements

TRAINING, SKILLS, AND INTERESTS

Programming, Analysis, and Research

- Python (Hugging Face, Pandas, NumPy, PyTorch, spaCy); R (data wrangling, regression, Bayesian modeling); JavaScript (experiment design)
- Experimentation: eye-tracking, self-paced reading (ExperimentBuilder, DataViewer, E-Prime, OpenSesame); survey design (Qualtrics); structured interviews

Academic and Professional Training

- Selected Ph.D. courses: Univariate & Multivariate Statistical Modeling in R; Computer Programming for Speech and Language Processing; Pragmatics; Doing Research in NLP; Accelerated NLP; Case Studies in AI Ethics; Text Technologies for Data Science; Machine Learning Practical

- UX design (online course): user research, wireframing, and usability-testing fundamentals, applied to conversational-AI co-design planning
- Interior architecture and design (2-year certified vocational training): human-centered spatial design and modeling, informing an applied, participatory approach to designing spaces and experiences for end users
- Languages: English, Persian, Azerbaijani, Turkish (+ basic French, German, Suret, Arabic)

Additional

- Trauma care psychology: corrective exercises for post-traumatic stress care

AWARDS AND SCHOLARSHIPS

- UKRI Centre for Doctoral Training in NLP Scholarship, University of Edinburgh, 2021–2025
- Exceptional Talents Award for Higher Education Programs Enrollment, Shahid Chamran University, Iran, 2010–2013
- Undergraduate Scholarship (National University Entrance Exam), Shahid Chamran University, Iran, 2006–2010

INVITED TALKS

- CDTalks, CDT in NLP: Language Model and Human Handling of Written Disfluency and Nonliteral Meaning, Edinburgh, 2024
- Brain, Cognition, and Development Seminar: Structural and Semantic Predictors of Written Sarcasm in Online Settings, Edinburgh, 2024
- CDTalks, CDT in NLP: Differences in Human and Machine Interpretation of Nonliteral Meanings, Edinburgh, 2023
- Psycholinguistics Coffee, School of PPLS: When Speech Becomes Writing: The Case of Disfluencies, Edinburgh, 2023
- Alan Turing Institute Student Networking Session: Tracing Sarcastic Meanings of Fillers on Twitter, London, 2022

WORKSHOPS, CONFERENCES, AND SERVICE

Attended / Presented

- UK and Ireland Speech (UKIS) Workshop & Speech Processing for Health Tutorial, London (2026) — Attendee/Trainee
- British Psychological Society – Scottish Branch, Perth (2025) — Speaker
- UKRI AI CDTs in NLP Joint Partners’ Summit, Edinburgh (2025) — Poster presenter
- SemDial (TrentoLogue), Rovereto (2024) — Speaker
- CLASP Conference on Learning with Small Data, Gothenburg (2023) — Poster presenter
- SemDial (DubDial), Dublin, Ireland (2022) — Poster presenter
- National Congress of Family Psychology Association of Iran, Ahvaz (2021) — Speaker
- Other trainee/attendee activities: Machine Learning in Cognitive Sciences, Tabriz (2021); Cognitive Rehabilitation, Tehran (2020); EEG Biofeedback, Urmia (2019); Premarital Counseling, Tabriz (2018); Transdiagnostic Treatments for Anxiety Disorders, Urmia (2018); Literary Criticism Conference, Tehran (2011); Biennial Congress on Psychology and Behavioral Sciences, Tabriz (2019)

Volunteering and Community Engagement

- Conference logistics volunteer, Architecture and Mechanisms for Language Processing (AMLaP), Edinburgh, 2024
- Recruitment volunteer, SEKHAVAT Coordination Centre (underrepresented gifted students’ funds), Urmia, 2018–2021
- Emotional support volunteer, cancer patients, Omid Hospital, Urmia, 2013–2015
- English language instructor, pediatric cancer patients, Shafa Hospital, Ahvaz, 2007
- Student assistant, psychology support team for runaway girls (sexual abuse/assault survivors), 2006–2007

REFERENCES

- Prof. Martin Corley, School of PPLS, University of Edinburgh — Martin.Corley@ed.ac.uk
- Prof. Hannah Rohde, School of PPLS, University of Edinburgh — Hannah.Rohde@ed.ac.uk
- Dr. Patrick Sturt, School of PPLS, University of Edinburgh — Patrick.Sturt@ed.ac.uk
- Prof. Daniel Mirman, School of PPLS, University of Edinburgh — daniel.mirman@ed.ac.uk
- Dr. Mostafa Zarean, Department of Psychology, University of Tabriz — zarean@tabrizu.ac.ir